

CAMPUS CARBON

Universal Campus Carbon Footprint Assessment & Management Platform (v2.0)

Organization:	Christ University	Reporting Year:	2025
Campus Name:	Christ University – Kengeri Campus	Assessment Status:	Completed
Campus Type:	University / College	Total Population:	6,450 occupants
Location:	Bengaluru, Karnataka, India	Built-up Area:	174,750.0 m ² (78.5 Acres)
Standard / Protocol:	GHG Protocol Corporate Standard / ISO 14064-1 (India CEA v19 Grid Baseline)	Generated On:	23 September 2026, 22:47

1. GROSS / POSITIVE EMISSIONS 1,796.54 tCO ₂ e	2. ELIGIBLE REDUCTIONS & REMOVALS -640.00 tCO ₂ e	3. NET CARBON FOOTPRINT 1,156.54 tCO ₂ e
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1. Executive Carbon Accounting Breakdown

Scope Classification	Activity Description	Emissions (tCO ₂ e)	Share of Gross (%)
Scope 1 (Direct)	Stationary DG diesel, canteen LPG, campus fleet combustion, refrigerants	338.20	18.8%
Scope 2 (Indirect - Electricity)	Purchased grid electricity & EV charging consumption	1,324.60	73.7%
Scope 3 (Other Indirect)	Commuting, solid waste disposal, freshwater supply, wastewater STP	133.75	7.4%
Total Gross Emissions	All combined operational activity emissions	1,796.54	100.0%
Eligible Removals / Reductions	Tree carbon sequestration, verified EV savings, waste diversion	-640.00	-35.6%
Official Net Carbon Footprint	Gross Emissions – Eligible Reductions & Removals	1,156.54	64.4%

Gross Intensity / Person: 0.2785 tCO ₂ e/person	Net Intensity / Person: 0.1793 tCO ₂ e/person	Gross Intensity / Area: 10.28 kgCO ₂ e/m ²	Renewable Energy Share: 28.0%	Waste Diversion Rate: 81.9%
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2. Building-wise Carbon Performance Leaderboard

Code	Building Name	Type	Area (m ²)	Occ.	Gross (t)	Red. (t)	Net (t)	tCO ₂ e/p	kg/m ²
DIN-01	Central Dining Hall & Foo...	Canteen / Kitchen	5,500	180	77.56	0.00	77.56	0.43	14.1
ENG-01	Engineering Block I (Civi...	Academic	14,500	600	0.00	0.00	0.00	0.00	0.0
ENG-02	Engineering Block II (CSE...	Academic	16,200	750	0.00	0.00	0.00	0.00	0.0
ARCH-01	School of Architecture & ...	Academic	9,800	400	0.00	0.00	0.00	0.00	0.0
MGMT-01	School of Business & Mana...	Academic	11,500	550	0.00	0.00	0.00	0.00	0.0
ARTS-01	Social Sciences & Humanit...	Academic	8,500	420	0.00	0.00	0.00	0.00	0.0
RES-01	Center for Research & Inn...	Laboratory	7,200	250	0.00	0.00	0.00	0.00	0.0
LIB-01	Central Knowledge & Libra...	Library	7,400	300	0.00	0.00	0.00	0.00	0.0
AUD-01	Main Auditorium & Cultura...	Auditorium	6,800	200	0.00	0.00	0.00	0.00	0.0
HST-01	St. Kuriakose Boys Hostel	Hostel	12,500	250	0.00	0.00	0.00	0.00	0.0

HST-02	Devadan Hall Boys Hostel	Hostel	11,800	220	0.00	0.00	0.00	0.00	0.0
HST-03	Carmel Hall Girls Hostel	Hostel	13,200	260	0.00	0.00	0.00	0.00	0.0
HST-04	Mother Teresa Girls Hoste...	Hostel	12,800	210	0.00	0.00	0.00	0.00	0.0
QTR-01	Staff & Faculty Residenti...	Staff Quarters	8,200	90	0.00	0.00	0.00	0.00	0.0
SPRT-01	Indoor Sports Complex & G...	Sports / Recreation	6,500	150	0.00	0.00	0.00	0.00	0.0
LAB-01	Advanced Computing & AI L...	Laboratory	6,200	220	0.00	0.00	0.00	0.00	0.0
LAB-02	Robotics & IoT Workshop	Laboratory	5,400	160	0.00	0.00	0.00	0.00	0.0
ADM-01	Central Administrative Bl...	Administrative	7,650	240	0.00	0.00	0.00	0.00	0.0
MED-01	Campus Health Center & Cl...	Hospital / Medical	7,200	100	0.00	0.00	0.00	0.00	0.0

3. Negative / Eligible Reduction Contributions Ledger

Category	Activity & Intervention	Methodology & Baseline	Status	Avoided / Sequestered
Solar Plant	Solar Clean Power Generation (Solar...	Grid Electricity Displacement (CEA India...	Eligible	-515.52 tCO2e
Gardening / Greenery	Biomass Sequestration by Campus Tre...	FSI Tree Biomass Growth Model & IPCC GPG...	Eligible	-61.60 tCO2e
Water Bodies	Ecological Wetland & Water Body Con...	Ramsar Wetland Sediment Carbon Stabiliza...	Eligible	-7.50 tCO2e
Waste Diversion	Avoided Landfill Methane via Compos...	Avoided Landfill Methane Generation (IPC...	Eligible	-36.08 tCO2e
Waste Diversion	Avoided Landfill Methane via Recycl...	Avoided Landfill Methane Generation (IPC...	Eligible	-19.30 tCO2e

4. Prioritized Recommendations & Decarbonization Action Plan

Priority	Category	Intervention Action Title	Intervention Type	Est. Impact
High	Electricity	On-site Rooftop Solar PV Expansion & Sma...	Emission Reduction	-25%
High	Electricity	HVAC & Lighting Deep Energy Efficiency R...	Emission Reduction	-15%
Medium	Greenery	Afforestation & Miyawaki Native Urban Fo...	Carbon Sequestration / Removal	-8%
Low	Water	100% Treated Wastewater Recycling for La...	Emission Reduction	-5%

5. Accounting Boundaries & Anti-Double-Counting Disclosures

Methodological Boundary: This assessment is executed under the operational control consolidation approach in strict accordance with the GHG Protocol Corporate Accounting and Reporting Standard and ISO 14064-1:2018 specifications.

Anti-Double-Counting Safeguards:

1. **On-site Solar Generation:** Solar energy consumed directly on campus displaces grid power and is reflected exclusively in reduced Scope 2 emissions. It is NOT subtracted a second time as a separate negative carbon offset.
2. **EV Fleet Accounting:** Operational EV charging electricity is quantified under Scope 2 electricity. Avoided transport reductions are calculated separately against an explicit ICE baseline.
3. **Carbon Sequestration:** Tree and vegetation biomass sequestration is modeled based on Forest Survey of India & IPCC GPG factors (22.0 kg CO2/tree/year) and reported in the eligible removals ledger.